Documentation Docker – Projet FitTrack

Eya Balti

ISIL 2-1

1. Contexte du projet :

Le projet FitTrack est une application mobile développée avec une architecture moderne basée sur des micro-services.
Afin de faciliter le déploiement, la portabilité et la reproductibilité de l'environnement, l'application a été conteneurisée dans Docker.

L'application est composée de trois services principaux orchestré a travers Docker Compose :

- **Base de données (db) : MySQL 8.4**
- **Backend (backend) : Spring Boot (Java 17)**

- **Frontend (frontend) : Expo Web (React Native)**

2. Architecture Docker du projet

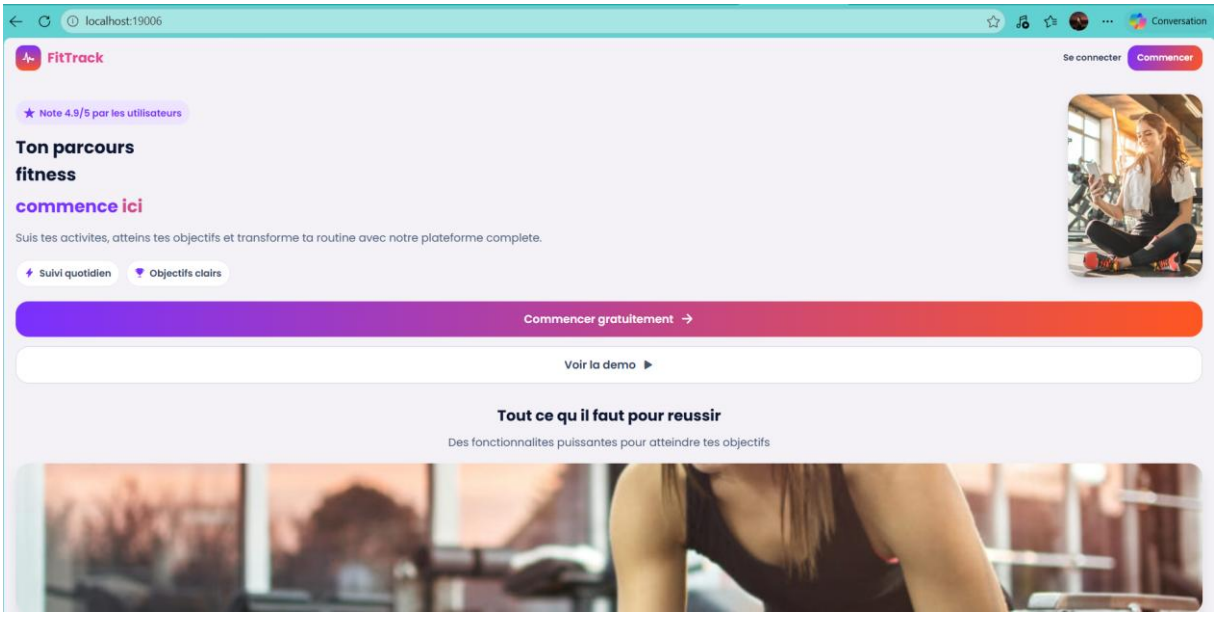
Services et conteneurs :

Service	Conteneur Docker	Description
db	fittrack-mysql	Base de données MySQL
backend	fittrack-backend	API REST Spring Boot
frontend	fittrack-mobile-web	Interface utilisateur

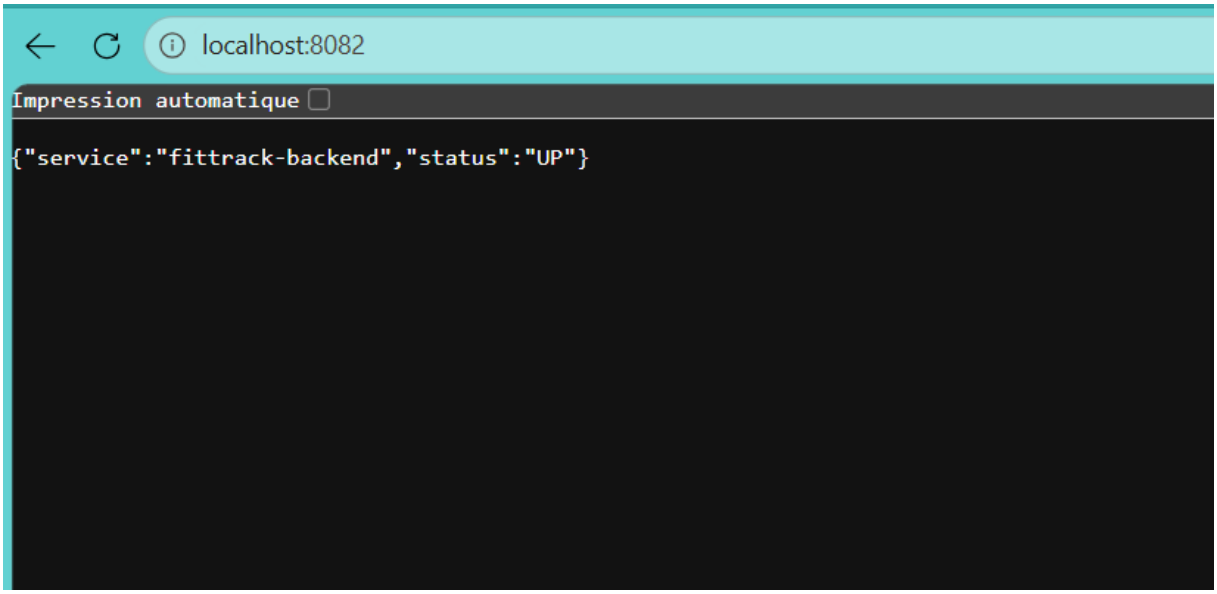
Ports utilisés

Service	URL / Port
Frontend	http://localhost:19006
Backend	http://localhost:8082
MySQL	localhost:3307

http://localhost:19006



http://localhost:8082



Configuration du fichier .env

Le fichier .env contient les variables d'environnement nécessaires au bon fonctionnement des services.

Configuration principale

- **BACKEND_PORT=8082**
- **FRONTEND_PORT=19006**
- **MYSQL_HOST_PORT=3307**
- **MYSQL_DATABASE=fittrack_db**
- **MYSQL_USER=fittrack**
- **MYSQL_PASSWORD=fittrack123**
- **MYSQL_ROOT_PASSWORD=root123**

Configuration du service mail

- **MAIL_HOST=smtp.ethereal.email**
- **MAIL_PORT=587**
- **MAIL_USERNAME=...**
- **MAIL_PASSWORD=...**
- **MAIL_FROM=...**

Étapes Docker réalisées

Dans cette section, nous présentons les étapes suivies pour déployer et tester l'application.

Ouverture du projet

Nous avons accédé au dossier du projet à l'aide du terminal :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

Étape 2 : Vérification de Docker et Docker Compose

Nous avons vérifié la disponibilité des outils Docker :

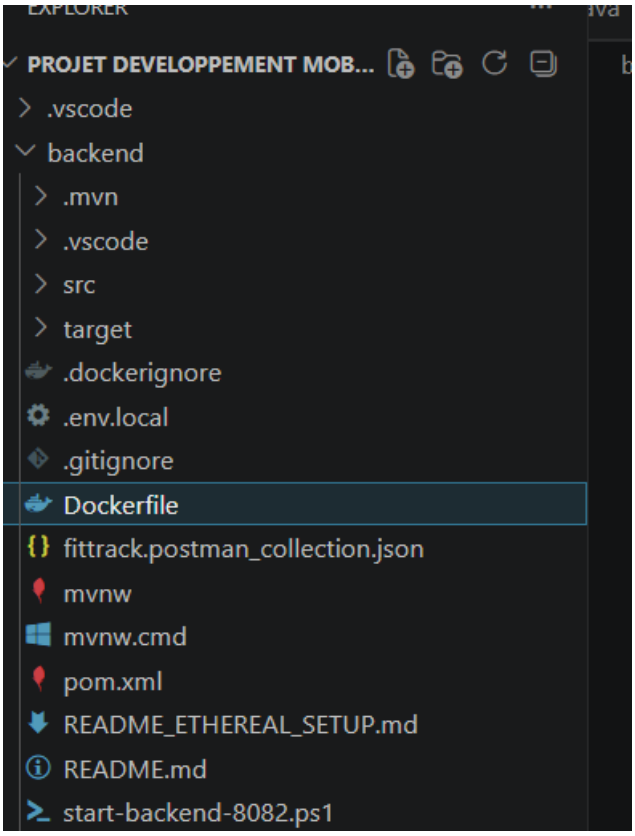
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Eya\Desktop\Projet developpement mobile> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose version
Docker Compose version v5.1.0
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

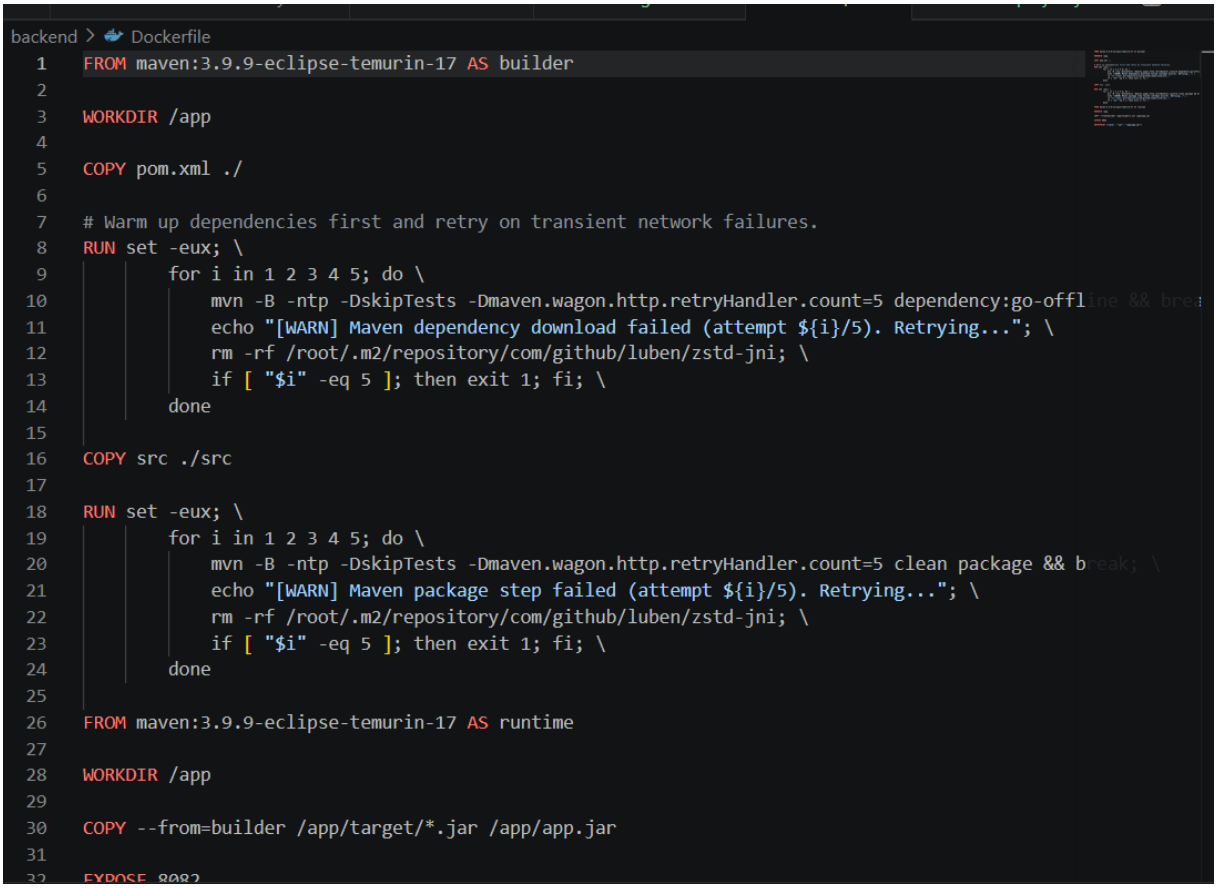
Création des fichiers Docker :

Création du Dockerfile du backend

Nous avons créé un fichier nommé Dockerfile dans le dossier backend.

Objectif : conteneuriser l'application Spring Boot.





backend > Dockerfile

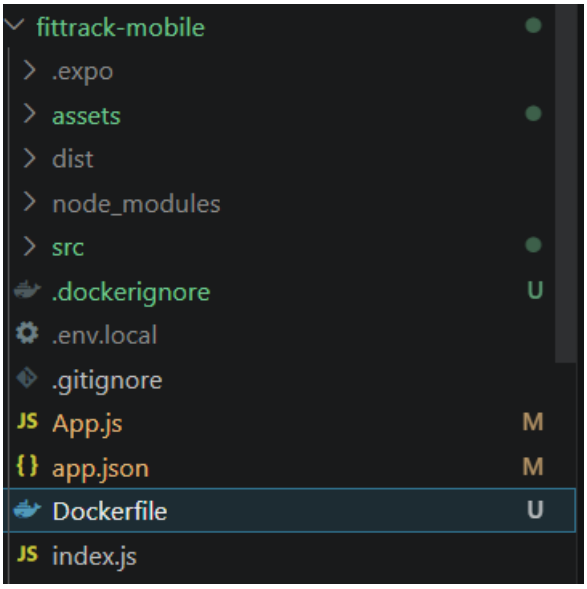
```
FROM maven:3.9.9-eclipse-temurin-17 AS builder

WORKDIR /app

COPY pom.xml ./

# Warm up dependencies first and retry on transient network failures.
RUN set -eux; \
        for i in 1 2 3 4 5; do \
            mvn -B -ntp -DskipTests -Dmaven.wagon.http.retryHandler.count=5 dependency:go-offline && break; \
            echo "[WARN] Maven dependency download failed (attempt ${i}/5). Retrying..."; \
            rm -rf /root/.m2/repository/com/github/luben/zstd-jni; \
            if [ "$i" -eq 5 ]; then exit 1; fi; \
        done

COPY src ./src

RUN set -eux; \
        for i in 1 2 3 4 5; do \
            mvn -B -ntp -DskipTests -Dmaven.wagon.http.retryHandler.count=5 clean package && break; \
            echo "[WARN] Maven package step failed (attempt ${i}/5). Retrying..."; \
            rm -rf /root/.m2/repository/com/github/luben/zstd-jni; \
            if [ "$i" -eq 5 ]; then exit 1; fi; \
        done

FROM maven:3.9.9-eclipse-temurin-17 AS runtime

WORKDIR /app

COPY --from=builder /app/target/*.jar /app/app.jar

EXPOSE 8082
```

Création du Dockerfile du frontend

Nous avons créé un Dockerfile pour l'application React Native (Expo Web).

Installation des dépendances

Lancement du serveur Expo

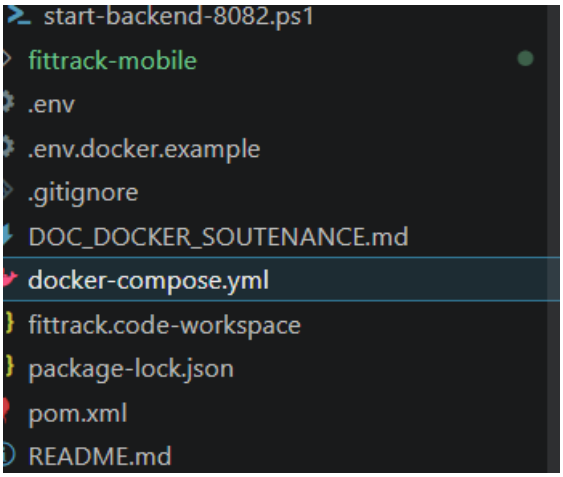


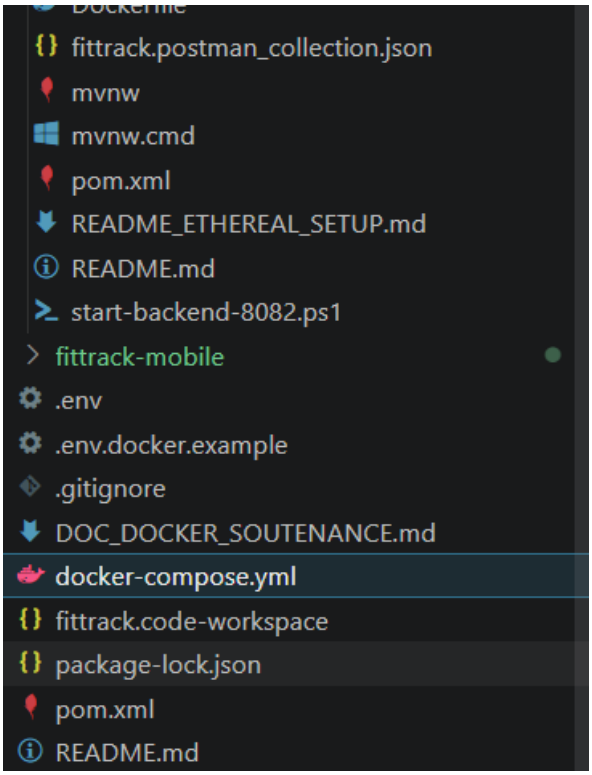
fittrack-mobile > Dockerfile

```
FROM node:20-alpine

WORKDIR /app

ENV CI=1

COPY package*.json ./
RUN npm install

EXPOSE 19006

CMD ["npx", "expo", "start", "--web", "--host", "lan", "--port", "19006"]
```

Création du fichier docker-compose.yml

Nous avons ensuite créé le fichier principal d'orchestration.

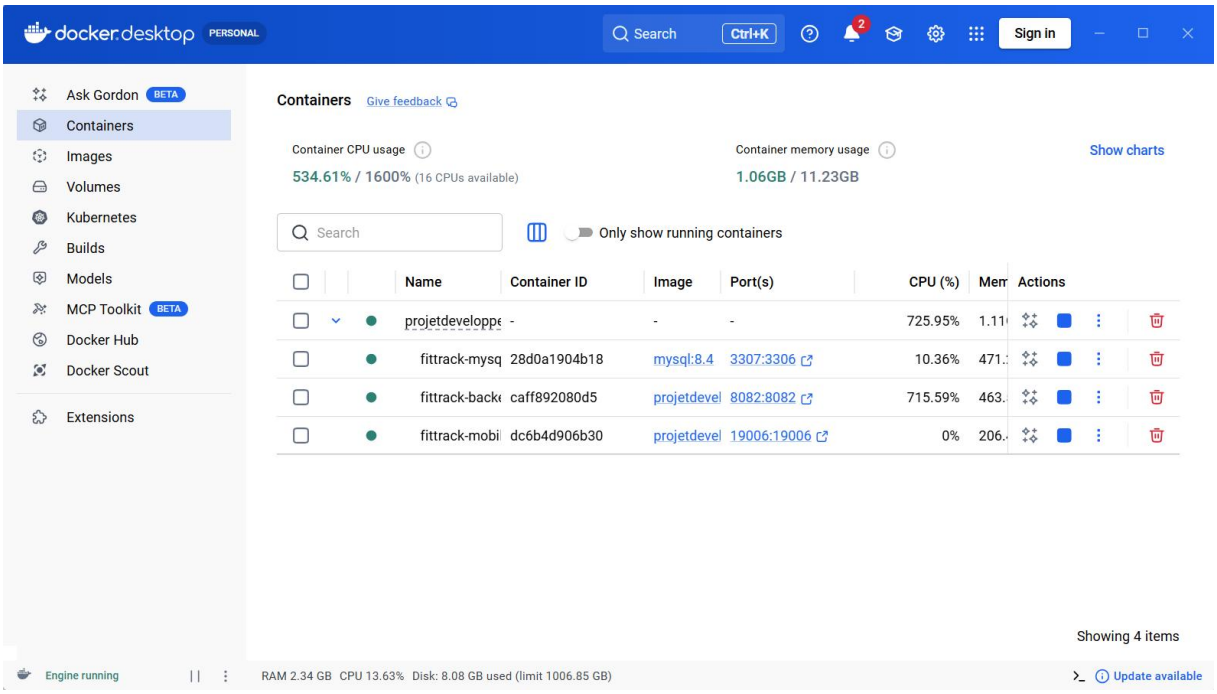

start-backend-8082.ps1
fittrack-mobile
.env
.env.docker.example
.gitignore
DOC_DOCKER_SOUTENANCE.md
docker-compose.yml
fittrack.code-workspace
package-lock.json
pom.xml
README.md


fittrack.postman_collection.json
mvnw
mvnw.cmd
pom.xml
README_ETHEREAL_SETUP.md
README.md
start-backend-8082.ps1
fittrack-mobile
.env
.env.docker.example
.gitignore
DOC_DOCKER_SOUTENANCE.md
docker-compose.yml
fittrack.code-workspace
package-lock.json
pom.xml
README.md

```yaml
docker-compose.yml
services:
  db:
    image: mysql:8.4
    container_name: fittrack-mysql
    restart: unless-stopped
    environment:
      MYSQL_DATABASE: ${MYSQL_DATABASE:-fittrack_db}
      MYSQL_USER: ${MYSQL_USER:-fittrack}
      MYSQL_PASSWORD: ${MYSQL_PASSWORD:-fittrack123}
      MYSQL_ROOT_PASSWORD: ${MYSQL_ROOT_PASSWORD:-root123}
    ports:
      - "${MYSQL_HOST_PORT:-3307}:3306"
    volumes:
      - mysql_data:/var/lib/mysql
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-u$$MYSQL_USER", "-p$$MYSQL_PASSWORD"]
      interval: 10s
      timeout: 5s
      retries: 20

  backend:
    build:
      context: ./backend
      dockerfile: Dockerfile
    container_name: fittrack-backend
    restart: unless-stopped
    depends_on:
      db:
        condition: service_healthy
    environment:
      SERVER_PORT: ${BACKEND_PORT:-8082}
      DB_URL: jdbc:mysql://db:3306/${MYSQL_DATABASE:-fittrack_db}?allowPublicKeyRetrieval=true&useSSL
```

Build et démarrage

```
PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose up -d --build
#22 naming to docker.io/library/projetdeveloppementmobile-backend:latest 0.0s done
#22 unpacking to docker.io/library/projetdeveloppementmobile-backend:latest 0.0s done
#22 DONE 0.5s

#23 [frontend] resolving provenance for metadata file
#23 DONE 0.1s

[+] up 5/5nd] resolving provenance for metadata file
 ✓ Image projetdeveloppementmobile-backend  Built                    5.1s
 ✓ Image projetdeveloppementmobile-frontend Built                    5.1s
 ✓ Container fittrack-mysql                 Healthy                  3.5s
 ✓ Container fittrack-backend               Started                  4.2s
 ✓ Container fittrack-mobile-web            Started                  3.6s
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```



Vérification des conteneurs

```
PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose ps
NAME                    IMAGE                                  COMMAND                  SERVICE    CREATE
D              STATUS                       PORTS
fittrack-backend        projetdeveloppementmobile-backend      "java -jar /app/app.…"   backend    About
a minute ago   Up About a minute            0.0.0.0:8082->8082/tcp, [::]:8082->8082/tcp
fittrack-mobile-web     projetdeveloppementmobile-frontend     "docker-entrypoint.s…"   frontend   About
a minute ago   Up About a minute            0.0.0.0:19006->19006/tcp, [::]:19006->19006/tcp
fittrack-mysql          mysql:8.4                              "docker-entrypoint.s…"   db         13 hou
rs ago         Up About an hour (healthy)   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

Vérification des images

```
PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose images
CONTAINER               REPOSITORY                         TAG              PLATFORM          IM
AGE ID            SIZE              CREATED
fittrack-backend        projetdeveloppementmobile-backend  latest           linux/amd64       49
2c99b0c469        278MB             13 hours ago
fittrack-mobile-web     projetdeveloppementmobile-frontend latest           linux/amd64       db
8cbd53ee76        249MB             14 hours ago
fittrack-mysql          mysql                              8.4              linux/amd64       da
906917ca4a        233MB             2 weeks ago
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

Logs backend

```
PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose logs --tail=40 backend
gerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
fittrack-backend  | 2026-04-09T11:53:30.970Z  INFO 1 --- [           main] eAuthenticationProviderMan
agerConfigurer : Global AuthenticationManager configured with AuthenticationProvider bean with name a
uthenticationProvider
fittrack-backend  | 2026-04-09T11:53:30.972Z  WARN 1 --- [           main] r$InitializeUserDetailsMan
agerConfigurer : Global AuthenticationManager configured with an AuthenticationProvider bean. UserDet
ailsService beans will not be used for username/password login. Consider removing the AuthenticationP
rovider bean. Alternatively, consider using the UserDetailsService in a manually instantiated DaoAuth
enticationProvider.
fittrack-backend  | 2026-04-09T11:53:32.675Z  INFO 1 --- [           main] o.s.b.w.embedded.tomcat.To
mcatWebServer  : Tomcat started on port 8082 (http) with context path '/'
fittrack-backend  | 2026-04-09T11:53:32.757Z  INFO 1 --- [           main] com.fittrack.FittrackAppli
cation         : Started FittrackApplication in 18.037 seconds (process running for 20.398)
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

Logs base de données

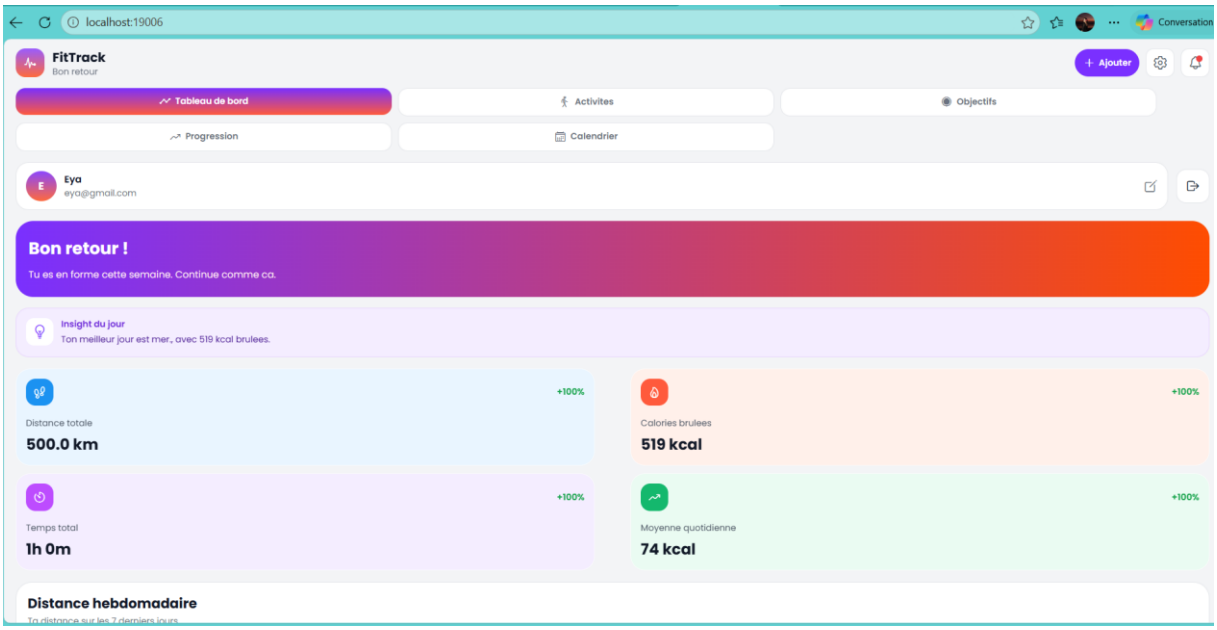
```
PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose logs --tail=20 db
as ended.
fittrack-mysql  | 2026-04-09T10:28:57.637177Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem
is self signed.
fittrack-mysql  | 2026-04-09T10:28:57.637291Z 0 [System] [MY-013602] [Server] Channel mysql_main conf
igured to support TLS. Encrypted connections are now supported for this channel.
fittrack-mysql  | 2026-04-09T10:28:57.646097Z 0 [Warning] [MY-011810] [Server] Insecure configuration
 for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choos
ing a different directory.
fittrack-mysql  | 2026-04-09T10:28:57.734101Z 0 [System] [MY-011323] [Server] X Plugin ready for conn
ections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
fittrack-mysql  | 2026-04-09T10:28:57.734506Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready
 for connections. Version: '8.4.8'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Communit
y Server - GPL.
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

Logs frontend

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Eya\Desktop\Projet developpement mobile> docker compose logs --tail=20 frontend
fittrack-mobile-web  | env: export EXPO_PUBLIC_API_LAN_HOST
fittrack-mobile-web  | Starting project at /app
fittrack-mobile-web  | Metro is running in CI mode, reloads are disabled. Remove CI=true to enable wa
tch mode.
fittrack-mobile-web  | Starting Metro Bundler
fittrack-mobile-web  | The following packages should be updated for best compatibility with the insta
lled expo version:
fittrack-mobile-web  |   @react-native-async-storage/async-storage@3.0.2 - expected version: 2.2.0
fittrack-mobile-web  |   expo-image-picker@55.0.16 - expected version: ~17.0.10
fittrack-mobile-web  | Your project may not work correctly until you install the expected versions of
 the packages.
fittrack-mobile-web  | Waiting on http://localhost:19006
fittrack-mobile-web  | Logs for your project will appear below.
PS C:\Users\Eya\Desktop\Projet developpement mobile>
```

Test

Accès depuis navigateur web



Accée depuis mobile

13:04

◀ Scanner de code

FitTrack
Bon retour

+ Ajouter

Tableau de bord | Activites | Objectifs

Progression | Calendrier

E
Eya
eya@gmail.com

Bon retour !

Tu es en forme cette semaine. Continue comme ca.

Insight du jour
Ton meilleur jour est mer., avec 519 kcal brulees.

+100%
Distance totale
500.0 km

+100%
Calories brulees
519 kcal

+100%
Temps total
1h 0m

+100%
Moyenne quotidienne
74 kcal

Distance hebdomadaire

Ta distance sur les 7 derniers jours